

Jesse Anderson

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Summary

Engineer with a Master's in Computer Science and a Bachelor's in Chemical Engineering, working across data systems, automation, and embedded hardware. Build ETL pipelines, internal tooling, and analytics that replace manual workflows, with a documented record of eliminating 900+ hours of repetitive engineering work per year. Co-author of six peer-reviewed publications.

Technical Skills

Languages: Python, Rust, C++, C#/.NET, TypeScript/JavaScript, MATLAB, SQL (Postgres/DuckDB), VBA, R, CUDA.
Data & ML: ETL, Pandas, Scikit-learn, PyTorch, OpenCV, statistical modeling, DBSCAN/HDBSCAN, Tableau, Power BI.
Engineering: Process automation, Aspen Plus, SuperPro, process control, blueprint and P&ID reading, AutoCAD, circuit design.
Infrastructure & Embedded: Docker, Kubernetes, AWS/GCP, Linux/Bash, CI/CD, React, Flask, WASM, ESP32, Raspberry Pi, MQTT.

Experience

Engineer | G5 Environmental | Chicago, IL September 2024–Present

- Architect and deploy embedded sensor modules (ESP32, pressure, RPM, GPS) to optimize fleet reliability.
- End-to-end ETL: ingest telemetry from the sensor network, transform via regression fits, load to an operator dashboard.
- Develop automated Python testing (pytest) and hardware-in-the-loop validation scripts to verify system behavior.
- Diagnose hydraulic leaks and electrical faults by analyzing sensor data logs, reducing diagnostic time by 40%.

Engineer | UL Solutions | Northbrook, IL June 2023–April 2024

- Design automation software in Python, VBA, and C#/.NET to eliminate 500+ hours of tedious workflows annually.
- Establish an ETL pipeline to reduce redundant analytical tasks by 400 hours per year.
- Develop Python software with OpenCV/PyMuPDF to automatically detect changes between CAD revisions.
- Determine project scope, plan of investigation, and specifications to initiate testing of products to UL standards.

Research Assistant | University of Illinois Chicago | Chicago, IL December 2020–June 2023

- Built MATLAB, R, and Python image analysis pipelines (DBSCAN, OPTICS, Ripley's K) over multi-gigabyte microscopy datasets.
- Co-authored six peer-reviewed papers (*Scientific Reports*, *Biophysical Journal*); cut a multi-hour analysis to seconds.

Engineering Intern | United Conveyor Corporation | Waukegan, IL May 2022–August 2022

- Pioneered software to automate 150 hours/yr of data entry and materials analysis to ASME, AWWA, and PPI standards.
- Piping and supports design work in AutoCAD; cut views from 3D CAD models to produce parts diagrams.

Peer Leader, Calculus-Based Physics | University of Illinois Chicago | Chicago, IL August 2020–December 2021

Tutored Physics I one-on-one and in groups; built an attendance analytics program measuring student attrition.

Safety/Project Manager | G5 Environmental | Chicago, IL June 2016–August 2019

Project lead at job sites: contract requirements met by CDL crews, repairs performed safely, parts sourced to avoid downtime.

Equipment Technician (Volunteer) | Remote Area Medical | Field Clinics July 2018–August 2018

Maintained generators and hydraulic/pneumatic dental tools; handled hazardous waste disposal at pop-up clinics.

Education

University of Illinois Urbana-Champaign | Master of Computer Science August 2025

GPA: 4.00 | Cloud Computing, Machine Learning, Distributed Systems, Scientific Visualization, Statistics, Databases

University of Illinois Chicago | BS Chemical Engineering May 2023

GPA: 3.77 | Concentrations: Biochemical Engineering & Process Automation | Minor: Mathematics & Computer Science

Senior Design: 60,000 ton/yr bacterial 1,3-propanediol plant simulation — 1st place, UIC ChemE Expo 2023

College of DuPage | Associate of Science | GPA: 3.65 Glen Ellyn, IL

Selected Projects

Engineering Tools Suite | tools.jesse-anderson.net — 40+ client-side engineering utilities in WebAssembly.

linreg-core Rust Crate | crates.io/crates/linreg-core — regression library with WASM, Python, and VBA bindings.

Oak Park Data Science Dashboard | opcrimeds.jesse-anderson.net — daily ETL and clustering of public police reports.

Certifications

Lean Six Sigma Yellow Belt (UL Solutions) | [Accelerated Computing with CUDA C/C++](#) (NVIDIA) | [Google Data Analytics](#) | [Google Project Management](#) | [Data Structures & Algorithms](#) (UC San Diego)